

A higher-resolution zoom into Stratum XIII of the parent atlas — the agent layer unfolded as ten sub-strata plus four wrapping meta-strata.
Stratum agenticum — denso compositum, ordo descendens, gradibus decem cum quattuor strata circumdantia

X S U B - S T R A T A · I V M E T A · D E S C E N D I N G			
X	END-USER · SURFACE form factors	chat · cli · ide · trigger · slack · whatsapp · voice · ar	10
I X	VERTICAL · PRODUCTS productized agents	sierra · decagon · glean · harvey · hippocratic · clay · rogo	9
V I I I	RUNTIME · SAFETY tool-boundary defense	lakera · nemo · llama-guard · bedrock · purview	8
V I I	EVAL · OBSERVABILITY trajectory analytics	braintrust · langsmith · langfuse · arize · galileo · inspect	7
V I	ACTION · SURFACES agent reaches into world	e2b · modal · browserbase · computer-use · vapi	6
V	PLANNING · REASONING cognition loop	react · planner-executor · o-series · extended-thinking	5
I V	MEMORY · STATE persistence	mem0 · letta · zep · graphrag · lab-native	4
I I I	TOOL · PROTOCOL interface to world	mcp · function calling · gateways · a2a	3
I I	AGENT · RUNTIMES harness	claude agent sdk · openai agents · adk · langgraph · mastra	2
I	FOUNDATION · CAPABILITY the agentic delta	claude · gpt-5 · gemini · deepseek · tau · swe · osworld	1
M E T A · W R A P P I N G F O R C E S			
A	CAPABILITY · SAFETY	asl · preparedness · frontier · aisi · metr	
B	REGULATION	eu ai act · ca sb 53 · sectoral · bartz · nyt	
C	ECONOMICS	per-task · outcome pricing · finops	
D	GEOPOLITICS	sovereign agents · export controls · jurisdiction	
K E Y	sub-stratum · component  meta-stratum — dependency		

Strata I–III · the model, the runtime, the protocol — the geological basement on which every agent runs.
Stratum primum, secundum, tertium · capacitas, harnesia, protocollum tooluse

III	TOOL · PROTOCOL	MODEL · CONTEXT · PROTOCOL · ADOPTION · FORK VECTORS		
		● MCP — Anthropic Nov '24 JSON-RPC; Py + TS SDKs ● Microsoft adoption — May '25 Copilot Studio · SK ● Registry · ~11.4K servers Apr '26 LF stats ● Pomerium identity proxy May '25 ● Stripe MCP first-party · canonical ● A2A protocol — Apr '25 agent-to-agent (Google)	● OpenAI adoption — Mar '25 Agents SDK · Responses API ● Hugging Face — Jun '25 MCP-compatible Spaces ● Cloudflare Workers AI MCP Dec '24 · gateway ● Function calling (legacy) OpenAI Jun '23 ● Linear · Cloudflare MCP first-party tier ● Fork vector — proprietary tool calls OAI latency wins	● Google adoption — Apr '25 Gemini · Vertex AI · A2A ● Linux Foundation — Dec '25 Agentic AI Foundation ● Kong MCP Gateway GA Jul '25 · F500 logos ● Structured outputs OpenAI Aug '24 ● GitHub MCP MSFT canonical ● Fork vector — quality split long tail abandonware
		REGISTRY · MAYHY'P2E6RSCALERS · CGOAMTMEIWTATYE DCATEGORY		
stratum tertium · the joint to the world		~11.4K4/4GA		
MCP · JSON - RPC · GATEWAYS				
II	RUNTIMES	AGENT RUNTIMES · HARNESSES · ORCHESTRATORS		
		● Claude Agent SDK Sep '25 · sub-agents · skills ● LangGraph 1.0 GA Oct '25 · production ● Pydantic AI typed validation · Logfire ● AutoGen / AG2 MSR + community fork ● Mistral Agents API May '25 · EU ● Aleph Alpha compliance-first · DE	● OpenAI Agents SDK Mar '25 · handoffs ● LangSmith obs layer · margin lives here ● CrewAI multi-agent · \$18M Oct '24 ● DSPy · BAML compile-time / IR layer ● Manus AI (CN) Butterfly Effect · viral '25	● Google ADK Apr '25 · Vertex Agent Engine ● Mastra TypeScript · YC W24 · Vercel ● Smolagents (HF) minimalist · Dec '24 ● Inspect AI (UK AISI) eval framework · OSS ● Sarvam AI (IN) Q1 '26
		SHIPPED · LASTLA1N4GGMROAPHARR · RUNTIMEGROSSMARG		
stratum secundum · the harness		4 SDKs\$40-60M (inf)THINNEST		
SDK · LOOP · SUB-AGENTS				
I	FOUNDATION · CAPABILITY	AGENTIC CAPABILITY DELTAS · NOT PROVIDER ECONOMICS		
		● Claude Opus 4.5 Nov '25 · 82.8% SWE-V ● ChatGPT Agent Operator+DR · OSWorld claim ● DeepSeek R1 / R2 open weights · mid-50s SWE ● Meta Llama 4 Apr '25 · -10-15 pts vs frontier ● Error compounding ceiling 0.95^20 ≈ 36% ● Tool use in thinking Claude 4.5 · GPT-5 · Deep Think	● Claude Sonnet 4.5 interactive executor ● Gemini 2.5 Pro Deep Think GAIA ~70 · BrowseComp ~50 ● Qwen QwQ / Qwen3 RLVR · open · CN ● Anthropic sub-agents fresh-ctx children · 6mo lead ● Computer-use ceiling OSWorld ~50% vs human 72%	● GPT-5 / 5.1 thinking ~80% SWE-V w/ tools ● Project Mariner GA Apr '26 · OSWorld ~38% ● Kimi K1.5 / K2 long-context CN · Moonshot ● Reasoning regression context degr. past 32K tokens ● Extended thinking 64K interactive vs back-office split
		SWE - BENCH · VERIOFSIWEODRLD (MAY 'R2E6A)SONING - MODEL TI		
stratum primum · the agentic delta		82.8%~50%9		
TAU · SWE · OSWORLD · BROWSECOMP				
SUBSTRATE · VOL III · THE AGENT LAYER				
Compiled for A. Yedi · Cycle MMXXVI · Rev. I				
PLATE II OF VI				

Strata IV-V · the inner cognition of the working agent – what it remembers and how it plans the next step.
Stratum quartum et quintum · memoria et ratio cinatio · circuitus interni

V	WHAT SURVIVED · TEST-TIME COMPUTE · ECONOMICS		
	● ReAct (Yao Oct '22) still dominant 2026 loop	● Plan-and-Solve (May '23) prompting · alive	● Tree-of-Thoughts (May '23) lab curiosity · rare prod
	● Reflexion (Mar '23) verifier sub-agent · alive	● AutoGPT / BabyAGI dead in prod · idea survived	● Planner-executor split durable architectural idea
PLANNING · REASONING	● Meta-planning / hierarchical emerging 2025-26	● o1 / o3 (Sep–Dec '24) reasoning model tier opens	● Claude extended thinking Feb '25 · 64K budgets
	● Gemini Deep Think (May '25) super-linear on hard	● DeepSeek R1 / R1-Zero Jan '25 · pure RLVR	● Qwen QwQ-32B · Kimi K1.5 open reasoning
	● Tool use IN thinking outer ReAct collapses	● `reasoning_effort` knob OpenAI · buyer-controlled	● PRM (Lightman May '23) in-model · not exposed
IV	● GRPO at trajectory DeepSeek · default RL		
stratum quintum · the inner loop	C X T R A J E C T O R Y	C · O D T E O K A E G N E S N T	T D O E K E E P N S R E S E A R C H
R E A C T · P L A N N E R - E X E C U T O R	4-25K	30-250K	0.5-5M
MEMORY · STATE	STANDALONE OR ABSORBED · COMPLIANCE WEDGE		
	● Mem0 (YC W24) \$24M A Dec '25 · 40K stars	● Letta (MemGPT) \$10M seed · A rumored '26	● Zep — Graphiti \$15M A Feb '25 · HIPAA
	● Cognee OSS pipeline · seed Dec '24	● Microsoft GraphRAG Apr '24 paper · Jul '24 OSS	● LightRAG (HKU) Oct '24 · cheaper graph
	● HippoRAG · PathRAG · OG-RAG research-only	● MemoryOS (Tsinghua) Sep '25 paper · CN	● ChatGPT memory Apr '24 paid · Sep '24 free
	● Claude Projects memory GA Apr '26	● Gemini personalization GA Mar '26	● Working memory (LangGraph) engineering not category
	● Procurement — Article 17 Mem0 forget API Feb '26	● HIPAA story (Zep) Athelas · F500 health	● Contextual integrity Nissenbaum · no vendor
	● Falsifiability test 70% pitches = RAG + writes		
stratum quartum · persistence	S T A N D A L O N E	C E I L L A I B N G M E (M I O N R F Y)	M A U C R U X # 5 (I N H E R I T E D
E P I S O D I C ·	\$200-400M	~800M	OPEN
	S E M A N T I C	P R O C E D U R A L	
S U B S T R A T E · V O L I I I · T H E A G E N T L A Y E R			
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Strata VI–VIII · where the agent reaches into the world, how we observe that it did the right thing, and what defends when it does not.
Stratum sextum, septimum, octavum · operatio, observatio, custodia

VIII	RUNTIME · SAFETY	GUARDRAILS · INJECTION DEFENSE · BUNDLING			
		● Lakera (Guard · Red) Series A '23 · B Q4 '25 ● HiddenLayer \$50M B Mar '24 ● Cranium KPMG spin · GRC-leaning ● Vertex AI Model Armor Cloud Next '24 ● Promptfoo (now OAI) red-team OSS · Sep '25 ● Constitutional classifiers Anthropic · baked in	● Protect AI · to Palo Alto ~\$700M Q3 '25 · the bell ● Apex Security Sequoia seed Mar '24 ● AWS Bedrock Guardrails Apr '24 · Auto Reasoning ● NeMo Guardrails 1.0 GA Apr '26 · OSS ● Garak (NVIDIA OSS) vulnerability scanner	● Robust Intelligence · to Cisco Aug '24 · AI Defense ● CalypsoAI \$23M B · federal angle ● Purview AI Hub · Defender MSFT Ignite Nov '24 ● Llama Guard 3 · Prompt Guard 2 Meta open weights ● EchoLeak CVE class Mar '26 · indirect PI	
		PURE - PLAYS AB SIONRDBIERDECTPI D E F E N S E S T A C K			
		stratum octavum · the tool boundary		18 MO UNSOLVED 5	
		PI · ACTIO N		G A T E S · S A N D B O X I N G	
VII	EVAL · OBSERVABILITY	AGENT EVAL · OBS · BENCHMARKS			
		● Braintrust \$36M A · B rumored '26 ● Arize Phoenix / AX \$70M C Dec '24 ● Patronus AI · Lynx · Glider \$17M A · defensible judge ● Helicone · proxy losing as native matures ● METR · time-horizon Mar '25 · cite not procure ● OSWorld · WebArena · GAIA computer/web/research	● LangSmith / LangChain \$25M C · LangGraph 1.0 ● Galileo \$45M B · C reported Apr '26 ● Comet Opik · DeepEval OSS · pytest-style ● Promptfoo (acq OAI) Sep '25 · neutrality gone ● TAU-bench / tau-2 Sierra-co · CX proxy	● Langfuse OSS · YC W23 · EU choice ● HumanLoop UK · FSI ● AgentOps CrewAI / AutoGen ecosystem ● Inspect AI (UK AISI) EU AI Act conformity ● SWE-Bench Verified · Live code agency · ~75-80%	
		O T E L G E N A I C O N C V O E N N S T O I L O I N D S A T I O N P (R 2 O 0 C 2 U 7 R) E M E N T			
		stratum septimum · trajectory analytics		JAN '26 GA PARTIAL OPEN	
		T R A J E C T O R Y		· R E P L A Y · O N L I N E	
VI	ACTION · SURFACES	SANDBOXED CODE · BROWSER · COMPUTER · VOICE OUTBOUND			
		● E2B OSS-core · ~\$15-25M ARR ● Vercel Sandbox GA Jan '26 · Firecracker ● Browserbase + Stagehand >\$50M ARR Q1 '26 ● Anchor · Hyperbrowser MCP-compatible 2025 ● Google Project Mariner GA Apr '26 · Gemini Ent ● LiveKit Agents WebRTC substrate ● Deepgram Nova-3 ASR · accent	● Modal agent-workload pivot · >\$50M ● Replit Agent sandbox bundled · prosumer ● Browserless · Skyvern incumbent · OSS ● Anthropic Computer Use Oct '24 · Claude for Chrome ● MSFT Copilot Vision Windows-native ● ElevenLabs Conv. AI voice up-stack ● Twilio · Telnyx · LiveKit Cloud telephony	● Daytona newer · price-aggressive ● HF Spaces · Codespaces ad-hoc · weaker fit ● Playwright + AI the boring baseline ● ChatGPT Agent Operator rebrand mid '25 ● Vapi · Retell · Bland voice trio · \$20M+ ARR ea ● Cartesia Sonic-2 ~40ms first-byte	
		V O I C E - T O - V O I C E S A F N L D O B O O R X T I M E L I C N U E P R O D U C T I O N			
		stratum sextum · agent reaches into world		500-800ms 18-24 MO NO	
		S A N D B O X ·		B R O W S E R · C O M P U T E R · V O I C E	
S U B S T R A T E · V O L I I I · T H E A G E N T L A Y E R					
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PLATE IV OF VI					

Strata IX—X · where enterprise revenue lands and where the human meets the agent — the productized application and its surface.
Stratum nonum · et decimum · productum · agenticum · et facies hominis

X	T W E L V E F O R M F A C T O R S · T H E B I N D I N G C O N S T R A I N T		
	● Chat — web/desktop ChatGPT 800M WAU · plateau	● Chat — mobile/on-device Apple Intelligence slot	● CLI / terminal Claude Code ~\$500M RR
	● In-IDE inline Cursor · Copilot · Windsurf	● Agentic trigger n8n · Zapier · cron · Art. 14	● In-browser sidebar Claude for Chrome · Comet 5M
	● Embedded in SaaS Copilot 365 ~30M seats	● Slack / Teams · headless Glean wedge	● WhatsApp · SMS · email Meta API Jan '26 · 2.9B
	● Voice — inbound Advanced Voice · AirPods	● Wearable · AR · ambient Ray-Ban >2M · Plaud >1M	● Computer use (UX) watched only · no reg ent
	● API · SDK · no-UI 2/3 OAI rev · majority Anthropic	● WhatsApp Business API Jan '26 · GLOBAL MAJORITY	● Coze on Feishu · WeChat CN parallel-track
	● Manus · Yellow.ai · Haptik non-US agent surfaces	● Pew chat-churn 41% Jan '26 · 'chat solved' = SF	● Anthropic mobile gap no AI slot · no voice
END-USER · SURFACES	M I C R O S O F T C O P I W L H O A T T S A A R P R P M A U B I N D I N G C O N S T R A I N T		
	\$5B+2.9BFORM		
IX	E N T E R P R I S E R E V E N U E L A Y E R · B U Y I N G M O T I O N B Y D O M A I N		
	● Sierra — \$175M+ ARR \$10B (Mar '26) · NYC dual	● Decagon — \$80M+ \$4.5B Sep '25 · NYC/SF	● Glean — \$300M+ \$7.2B Sep '25 · NYC growing
	● Harvey — \$100M+ \$5B Feb '26 · NYC office	● Hippocratic — \$50M+ \$2B Feb '26 · NYC HQ	● Augment Code — \$40M \$977M Nov '25 · enterprise
	● Cursor — \$500M RR \$9.9B Jul '25 · IDE leader	● Cognition (Devin) \$3B Codeium absorbed Mar '26	● Replit Agent — \$150M \$3B Sep '25 · prosumer
	● Lovable — \$80M Q1'26 fastest EU SaaS · Stockholm	● Abridge — \$120M+ \$2.75B Jun '25 · scribe	● Hebbia — \$50M+ NYC HQ · ex-2Sigma · raise
	● Clay — \$80M+ NYC HQ · \$1.5B Jan '26	● Rogo — \$30M+ NYC HQ · \$400M Jan '26	● Ramp AI org 13B·NYC·20B raise rumor
	● Runway — \$100M+ NYC Chelsea · creative	● ElevenLabs — \$200M+ voice platform · \$3.3B	● Cresta · Ada · Intercom Fin CX maturing/stalling
	● Parloa · PolyAI · Regal EU/UK + NYC sales voice	● EvenUp · Eve · Spellbook legal long tail	● 11x · Artisan · AiSDR synthetic SDR · 11x flat
	● Mistral Le Chat Ent. €600M Mar '26 · EU sovereign		
VERTICAL · PRODUCTS	W I N N E R - T A K E - M O B S E T T S # E 2 G M N E Y N C T S T A R O G U E T T C S O M E P R I C I N G (T		
	3/910+CX ONLY		
stratum decimum · form factor B U Y E R · T H R E A T · A D O P T I O N	L E G A L · H E A L T H · F I N A N C E		
stratum nonum · productized agents C X · C O D E ·	L E G A L · H E A L T H · F I N A N C E		

A N A T L A S O F T H E A G E N T S T R A T A			VI
META · WRAPPING FORCES			PLATE VI OF VI
Meta A–D · the four forces that wrap the agent layer: capability · safety regimes, regulation, economics, geopolitics. Quattuor stratorum meta · circumscribunt agenticam stratam			
D GEOPOLITICS	WHERE AGENTS RUN · EXPORT CONTROLS · SIT - CONTROLS		
	- Mistral Le Chat Ent. €600M @ €11B Mar '26 - Sarvam AI (IN) \$41M · Q1 '26 raise rumor - Coze on Feishu ByteDance · parallel-track - Agent-capability export rule 2026–27 regulatory front	- G42 · Falcon · UAE Humain ties · sovereign AI - Krutrim (Ola IN) \$50M · political tailwind - WeChat Mini-Programs agentic Q2 '26 - EU residency lever AI Act + GDPR + Schrems II	- Humain (KSA) MI300/350 commits · \$10B - Manus (CN) Butterfly · \$500M rumor - US Commerce auth — Nov '25 G42 + Humain chips - Trajectory siting where memory + actions reside
	meta · sovereign agents S O V E R E I G N · E X P O R T · J U R I S D I C T I O N		
C ECONOMICS	FROM PER-TOKEN TO PER-TASK · OUTCOME PRICING		
	- Token price · −4× / −10× Q1 '24 to Q1 '26 - Deep research · 0.5–5M tok ~1–20 / task - AWS Bedrock · OpenRouter tier routing standard - Per-seat survivors code · knowledge · finance	- CX trajectory · 4–25K tok ~0.02–0.30 / task - Planner-executor split 70–85% on cheap tier - Outcome pricing — CX per-resolution · \$1–4 (Taylor) - FinOps for Tokens (Bet #4) 12–18 mo window	- Code agent · 30–250K tok ~0.20–3 / task - Vercel AI Gateway routing first-class · GA '26 - Outcome — healthcare per-encounter · per-call \$9/hr - Hyperscaler bundling Bedrock auto-optim threat
	meta · the inversion P E R - T A S K · O U T C O M E · F I N O P S		
B REGULATION	EU ARTICLE 14 · CA SB 53 · BARTZ · SECTORAL		
	- EU AI Act · Art. 55 GPAI enforceable Aug 2 '25 - CA SB 53 frontier disclosure effective Jan 1 '26 - Bartz v. Anthropic \$1.5B training-data settle - HIPAA · FDA SaMD health agent fence	- EU AI Act · Art. 14 oversight draft Apr '26 · agentic - NY S5641 · state-level advancing 2026 - NYT v. OpenAI vendor terms repricing - TCPA · state consent voice agent fence	- EU AI Act · conformity drafts Feb '26 · Inspect AI cited - Trump Dec '25 preemption EO courts through '26-'27 - FINRA · OCC · BD rules finance agent fence - EU AI Office structural regulator
	meta · the gauntlet E U A I A C T · C A S B 5 3 · S E C T O R A L		
A CAPABILITY · SAFETY	LAB-SIDE SAFETY REGIMES · EVAL INSTITUTES		
	- Anthropic ASL framework ASL-3 Opus 4 May '25 - Preparedness adjustment clause Risk #3 · open question - UK AISI Nov '23 · Inspect framework - EU AI Office regulatory counterpart	- Anthropic RSP — Nov '24 binding policy - Google Frontier Safety May '24 · maturing - US AISI (post-restruct) early '26 · narrower scope - ASL-4 capability emergence 2026–27 question	- OpenAI Preparedness Framework tracked categories - DeepMind capability evals auto-replication · cyber - METR (research org) time-horizon · 7-mo doubling - Model cards for agents in-flight standardization
	meta · the deploy gate A S L · P R E P A R E D N E S S · F R O N T I E R		
S U B S T R A T E · V O L I I I · T H E A G E N T L A Y E R			
Compiled for A. Yedi · Cycle MMXXVI · Rev. I			PLATE VI OF VI